Visal Chathuranga

AI/ML Engineer · Full Stack Developer

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Professional Summary

AI/ML Engineer with hands on experience across **20+ AI and full stack projects**. These include agent systems that call real APIs, RAG document platforms, HL7 and EDI healthcare integrations, and multi tenant SaaS running in production. B.Sc. in Electrical & Electronic Engineering from the University of Peradeniya with *Second Class Upper Division Honours*. Currently building AI healthcare products at MedcubeUSA using FastAPI, Next.js and cloud services.

Technical Skills

Languages	Python ▪ TypeScript ▪ JavaScript ▪ SQL ▪ Bash / PowerShell ▪ HTML ▪ CSS
Frameworks	Django / DRF ▪ FastAPI ▪ Next.js ▪ Node.js ▪ Flask ▪ Express ▪ React ▪ Electron
AI / ML Libraries	TensorFlow ▪ Keras ▪ PyTorch ▪ Scikit-learn ▪ HuggingFace ▪ LangChain ▪ LangGraph ▪ fastembed ▪ OpenCV ▪ MediaPipe ▪ Albumentations ▪ timm ▪ LightGBM ▪ NLTK
AI Engineering	RAG ▪ Hybrid Search (BM25 + RRF) ▪ Multi Agent Orchestration ▪ MCP ▪ Vercel AI SDK ▪ OpenAI Assistants & Realtime APIs ▪ Vision OCR ▪ Prompt Engineering
Data & Libraries	Pandas ▪ NumPy ▪ Matplotlib ▪ Seaborn ▪ OpenAI SDK ▪ Prisma ▪ Drizzle ORM ▪ SQLAlchemy ▪ PDF.js ▪ MiniSearch ▪ jQuery
Databases	PostgreSQL ▪ MySQL ▪ SQL Server ▪ MongoDB ▪ Pinecone ▪ Qdrant ▪ ChromaDB ▪ FAISS ▪ Neo4j
Cloud & DevOps	AWS (S3 ▪ Textract ▪ Transcribe Medical ▪ EC2) ▪ Vercel ▪ Docker ▪ Kubernetes ▪ Git ▪ GitHub Actions ▪ CI/CD ▪ PM2 ▪ DVC ▪ MLflow
Healthcare & EDI	HL7 v2.3 ▪ X12 EDI 835 ERA ▪ CARC / RARC Codes ▪ NDC & RxNorm ▪ EHR Integration ▪ PHI Handling
Integrations	OpenAI API ▪ Claude API ▪ Gemini API ▪ Twilio (Voice & WhatsApp) ▪ Stripe ▪ QuickBooks ▪ Playwright ▪ JWT ▪ WebSocket ▪ SSE ▪ REST APIs

Experience

Sep 2025 – Present

Software Engineer (AI/ML) | *MedcubeUSA*

Colombo

- Designed and built production AI systems for healthcare clients, covering agent based record search, RAG document platforms, voice assistants and automated document processing, using OpenAI models, AWS services and vector databases.
- Owned the healthcare data integrations that downstream teams depend on, including HL7 v2 prescription feeds into multi tenant EHR databases and X12 EDI 835 remittance ingestion, both running as continuous production services.
- Delivered full stack SaaS platforms end to end, taking each product from requirements and database design through REST APIs, user interface, cloud deployment and post release support.
- Mentored two interns, setting their objectives, reviewing their code and providing daily technical guidance.
- Collaborated across the full delivery team, covering **product management, business analysis, quality assurance and engineering**, contributing to requirement reviews, technical design decisions and defect resolution through the team's tracked workflow.

Jul 2024 – Oct 2024

AI/ML Engineering Intern | *GPV Lanka (PVT) Ltd*

Negombo, Sri Lanka

- Built AI powered desktop applications for reflow oven temperature prediction and feeder management using Flask, TensorFlow, Keras, Electron.js and Python.

Jul 2023 – Oct 2023

Automation Engineering Intern | *Phoenix Industries Ltd*

Makandura, Sri Lanka

- Engineered an automated toothbrush packing machine and a defect detection system for Astra Cups, improving production efficiency.

Projects

GitHub

Multi Tenant WhatsApp Petty Cash Management Platform

Stack: Django 5.2 ▪ DRF ▪ PostgreSQL ▪ Webix Jet ▪ OpenWA ▪ AWS Textract ▪ django-q2 ▪ JWT ▪ SSE ▪ Playwright

- Built multi tenant WhatsApp and web expense claims with Textract OCR, receipt handling and multi level approvals.
- Engineered Django/DRF APIs with PostgreSQL, JWT/RBAC, audit logs, signed files and real time SSE updates.
- Added duplicate receipt checks, GPS travel claims, cash advances and appeals directly through WhatsApp.

GitHub

APEX Team: Self Coordinating 14 Agent AI Development System

Stack: Python ▪ FastAPI ▪ Next.js 14 ▪ TypeScript ▪ MCP (81 tools) ▪ WebSocket ▪ Playwright ▪ MITRE ATT&CK ▪ NIST CSF 2.0

- Built an AI development team that starts **14 specialist agents** from one command, covering architecture, backend, frontend, database, testing, security and DevOps roles. All agents share one file locked state file, tested at 200 concurrent writes with no data loss.
- Created a single **MCP server with 81 tools** that budgets tokens, shares summaries between agents and routes work across **2,195 skills**.

Hash based prompt caching cut token use by **30% to 70%**.

- Added a security suite of three agents and a six stage debate engine where agents propose, critique, revise, vote, score and judge. Shipped a live dashboard over WebSocket and one click installers for Windows, macOS and Linux.

GitHub

Agent Cortex: Self Hosted Long Term Memory Service for AI Agents

Stack: Python ▪ FastAPI ▪ Qdrant ▪ fastembed (ONNX) ▪ MCP ▪ BM25 ▪ Docker ▪ Caddy ▪ pytest

- Built and open sourced a memory server that lets AI agents remember things between sessions. It exposes **22 MCP tools, 49 REST endpoints and 35 CLI commands** over a plain Markdown folder, with Qdrant as a rebuildable search index.
- Combined vector search with a keyword ranker I wrote from scratch and merged both result lists using Reciprocal Rank Fusion, then reordered results by how useful and how often used each memory was.
- Packed each recall response into a token budget across **8 layers** and made the critical layers impossible to drop. Added a fact store that keeps the old value when a fact changes. Shipped as a pip package and Docker image with **91 offline tests**.

AeliusCase Global Search: Natural Language Case Search Agent

Stack: Next.js 15 ▪ React 19 ▪ TypeScript ▪ Vercel AI SDK ▪ OpenAI GPT-4o mini ▪ MongoDB ▪ OpenAI Vector Stores ▪ Vitest

- Built an AI agent that answers plain English legal questions by calling the right APIs itself, choosing from a **22 tool registry**, running a multi step tool loop capped at 5 steps, and streaming each answer back token by token.
- Built a **three stage router** that narrows the tool list before the model runs. A regex intent classifier and 8 domain modules go first, then an LLM classifier confirms the domain, so the agent is shown at most 8 tools per request instead of all 22.
- Stopped the agent inventing answers. Dates, staff roles and result counts are computed on the server and injected as verified facts, a whitelist blocks invented filters, and a loop guard forces an answer when the agent repeats the same tool call.
- Added **RAG** over the product guide with OpenAI Vector Stores, scoring retrieved chunks and passing only the top 3 into the prompt. Rebuilt context handling after one live request reached **186K tokens** against a 128K limit, trimming old tool output and keeping personal identifiers out of the model entirely. Covered with **539 tests**.

GitHub

VScanMail: AI Powered Digital Mail Custodian SaaS

Stack: Next.js 15 ▪ TypeScript ▪ MySQL ▪ Drizzle ORM ▪ OpenAI GPT-4o Vision ▪ Stripe ▪ QuickBooks ▪ AES-256-GCM ▪ TOTP

- Built a multi tenant SaaS for scanned mail and cheques where every client gets its own database table, so records can never leak between clients. Admin views join across all tenant tables at query time, even when older tables have fewer columns. Runs **144 endpoints**.
- Used GPT-4o Vision to read cheques, spot opened envelopes and flag returned cheques. Kept the final decisions in plain code, so amount checks, name matching and date checks never depend on what the model thinks.
- Encrypted bank account numbers with **AES-256-GCM** and added a hashed index so an account can be found without decrypting it. Added app based two factor login and an audit log of **47 actions** that stores the state before and after each change.
- Added Stripe subscriptions with a six state overdue flow, usage quotas that block scanning at the plan limit, and a manual payment path for clients billed offline.

GitHub Demo

DocSearch AI: Legal Document Intelligence Platform

Stack: Node.js ▪ TypeScript ▪ PostgreSQL ▪ MongoDB ▪ Pinecone ▪ MiniSearch ▪ OpenAI GPT-4o mini ▪ PDF.js ▪ SSE ▪ Playwright

- Rebuilt search so it uses both meaning and exact words. Pinecone vector results and a local keyword index are merged with Reciprocal Rank Fusion, with a SQL phrase search as a final fallback.
- Made every citation provable. The model has to quote the source text word for word, and if that quote is not in the retrieved pages the citation is removed. A Playwright test keeps this locked in.
- Highlighted the exact answer inside the PDF viewer by saving text position ranges alongside each vector. Replaced Tesseract with GPT-4o mini vision, which only runs on pages that look like scans, so cost stays low.
- Cut upload time from around 4 minutes for a 30 page file by batching 100 chunks per embedding call, and added question rewriting so follow ups keep their context.

ClaudMD DxScript eRx: Multi Agent HL7 Prescription Integration Pipeline

Stack: Python 3.11 ▪ LangGraph ▪ hl7apy ▪ SQLAlchemy ▪ SQL Server ▪ pyodbc ▪ Pydantic ▪ Windows Service

- Built a **three agent LangGraph system** that reads HL7 prescription files dropped by an eRx vendor and writes each one into the correct clinic database. Handled 14,162 messages across 6 clinic sites.
- **Agent 1, Ingestion:** scans the shared folder, waits until each file is fully written, parses the HL7 message and saves it as JSON with a hash, so no file is processed twice.
- **Agent 2, Routing:** walks a **seven step chain** of clinic, connection, patient, visit, order, drug code and prescription line. It reads the clinic ID from inside the message and builds that clinic's database connection at run time, even when clinic databases differ in columns.
- **Agent 3, Monitoring:** runs both inside the pipeline and on its own thread, so each side checks that the other is still alive. It marks the system healthy, failed, hung or stale and records only the moments the state actually changes.
- Made writes safe by requiring two separate flags before saving, never running updates or deletes, and replacing a full admin database login with one limited to five tables. Cut a full folder scan from **341 to 40 seconds**.

EDI 835 ERA Converter and Payment Ingestion Platform

Stack: Python ▪ FastAPI ▪ Next.js 14 ▪ MongoDB ▪ MySQL ▪ SQLAlchemy ▪ Paramiko SFTP ▪ JWT ▪ pandas

- Wrote an **X12 835 parser** from scratch that reads insurance payment files, handles 17 segment types, works out the file delimiters from the header, and explains every adjustment using **1,434 standard reason codes**.
- Built a service that pulls files from two clearing houses over SFTP every minute, looks up each claim in the billing system and stores the result. It has saved 11,634 payment records from 108 files and matched invoices for **98%** of them.

- Made sure no file is ever imported twice using a file tracker and a unique database index. Added a circuit breaker that pauses calls after five failures, plus a login protected lookup API with rate limits and audit logging.

Amazon Textract Invoice PDF Processor (Enhanced with LLM)

Textract Only +LLM

Stack: Python ▪ AWS Textract ▪ AWS S3 ▪ Gemini API ▪ JSON

- Built an invoice reading pipeline on AWS Textract with S3 storage, MD5 duplicate detection and table parsing, reaching **95% accuracy** and cutting manual work by **80%**.
- Added a Gemini layer that cleans up dates, currencies and payment terms, so the pipeline also handles invoice formats it has not seen before.

MedScribe Pro: Medical Transcription Platform

GitHub

Stack: FastAPI ▪ AWS Transcribe Medical ▪ AWS S3 ▪ OpenAI GPT-4 ▪ Python

- Built a FastAPI service that turns medical audio into speaker labelled transcripts (doctor, patient, nurse) with AWS Transcribe Medical, then writes clinical summaries with GPT-4.
- Sped it up with S3 transfer acceleration, multipart uploads and concurrent workers.

AI Voice Agent with RAG and Telephony

GitHub

Stack: FastAPI ▪ OpenAI Realtime API ▪ Twilio Voice ▪ Pinecone ▪ WebSocket ▪ Python

- Built a phone based voice assistant on the OpenAI Realtime API and Twilio, with live two way audio streaming for natural conversation.
- Added a Pinecone RAG pipeline with two modes: a full agent with web search and tools, and a documents only mode for regulated work.

Diabetic Retinopathy Detection: Deep Learning Classifier

Stack: Python ▪ PyTorch ▪ EfficientNetB3 ▪ timm ▪ Albumentations ▪ Scikit-learn

- Trained an EfficientNetB3 model to grade diabetic retinopathy into 5 severity levels, scoring **QWK 0.83** on APTOS 2019 and 0.57 on EyePACS across more than 35,000 images.
- Built a two phase training pipeline with mixed precision, weighted sampling, patient level splits and test time augmentation, plus auto resuming checkpoints for free Colab and Kaggle GPUs.

Education

B.Sc. Engineering (Hons.) in Electrical & Electronic Engineering — 2021 – 2025
GPA: 3.4 / 4.0

University of Peradeniya
Second Class Upper Division Honours

GCE Advanced Level — 2010 – 2018
Mathematics: A ▪ Chemistry: A ▪ Physics: B

Dharmasoka College, Ambalangoda

Certifications

IBM / Coursera — AI Engineering Specialization ▪ Deep Neural Networks with PyTorch (Honors) ▪ Building Deep Learning Models with TensorFlow ▪ AI Capstone Project with Deep Learning ▪ Introduction to Deep Learning & Neural Networks ▪ Introduction to Computer Vision & Image Processing ▪ Deep Learning and Reinforcement Learning ▪ Introduction to Web Development

DeepLearning.AI — NLP with Classification and Vector Spaces ▪ NLP with Probabilistic Models

Other Providers — Machine Learning with Python, Honors (Coursera) ▪ Generative AI Mastery with 15+ Real Time Projects (Udemy) ▪ Complete Java Full Stack Training (Digimax Education) ▪ Deep Learning Neural Networks & Image Processing (edXcope)

Extracurricular Activities

Treasurer **IEEE Microwave Theory and Techniques Society (MTTS)** — Jan 2024 – Aug 2025

Team Member **IEEE MTTS Programming Team** — Jan 2023 – Jan 2024

Member **Explorer's Club, University of Peradeniya** — 2022 – Present

Volunteer **Arunella Social Development Initiative** — 2022 – 2025
Supporting school students in underprivileged areas.

Soft Skills & Interests

Soft Skills — Adaptability ▪ Problem Solving ▪ Teamwork ▪ Communication ▪ Creativity ▪ Time Management

Interests — Exploring New Technologies ▪ Playing Chess ▪ Environmental Exploration ▪ Watching Films

Non Related Referees

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